H1B Lottery Registration Automation - PRD

Problem Statement

During every H1-B season, if there's a registration error, it implies that a client might not get their shot at the lottery. Attorneys at Alma are spending 10-15 minutes per beneficiary on work that is mostly clicking, copying, and praying they didn't make a typo. This PRD aims to eliminate that.

User Persona : *"Sarah is a senior immigration attorney. During H-1B season she handles 80-120 beneficiary registrations across 15-20 employer clients, all within a 14-day window. A single duplicate submission or data error can cost a client their lottery slot for the year."*

The worst part of the current process isn't the time, it's the uncertainty. An attorney uploads a CSV, gets a bunch of USCIS error messages they have to interpret manually, fix the data, re-upload, and still they aren't sure if the registration went through until they see a green banner. Then they do it again for the next client. During a 14-day window. With no margin for error.

Currently, attorneys perform all steps manually, spending 10-15 minutes per registration doing -

1. **Data Preparation:** Gathering information, validating data formats, checking for inconsistencies or missing data.

2. **Validation & Review:** Checking for duplicates, reviewing registration summaries, downloading confirmations.

3. **Payment & Submission:** Processing $215/beneficiary fees, submitting Form G-28 for legal representatives and waiting for approval.

Pain Points

1. **Manual Data Entry Errors:** Typos in passport numbers or other details, incorrect wage levels, mismatched SOC codes cause registration invalidation.

2. **Duplicates Detection:** USCIS tool only checks within single drafts, not across multiple uploads.

3. **Time-consuming and cumbersome :** Spending 10-15 mins per application adds up quickly for an entire batch and makes the entire process cumbersome.

Existing User Journey

H1B registration process

1. Logging into a company account.
2. Go to the 'File a Form' section and choose H-1B from the drop-down menu.
3. Data info preparation.
    a. Bulk upload
        i. Pre-upload - Preparing data in the template form and validating/ensuring that the data meets the standards.
    b. Individual upload
4. Security reminder page.
5. Upload, either the completed CSV or individual form entries.
    a. Bulk upload
    b. Individual entry upload - This is where most of the invisible work lives. The USCIS template has strict formatting rules: country codes, wage level codes, SOC codes. An attorney managing 80 beneficiaries is essentially doing data engineering. One misformatted field invalidates the row. There is no preview, no warnings before upload, no way to know if the data is clean until errors are flagged.
6. Review any errors. Ensure all errors are cleared. A green banner should appear when the data is accepted.
   USCIS returns error messages in unstructured pros, just paragraphs of text the attorney has to read and interpret. If there are errors, they go back to the spreadsheet, fix the data, re-upload, and wait again. For a batch of 80 beneficiaries this can happen multiple times. During a 14-day window.
7. Ensure all the errors are cleared and data is corrected, there should be a green banner shown to the user.
8. Click ‘Draft registration’ to view it, and if it was an individual data entry click, ‘Download csv of beneficiary table’ to keep a copy of it for user’s use**]**
9. Fill in the signatory's info.
10. User knows must select ‘click duplicates’ if they uploaded individual entry **-** Check for duplicates - if you entered info using template, it will have already checked for duplicates, if you haven’t or if you entered info directly into the form, you can use the ‘check for duplicates’ tool in the H1B registration process. Using the tool doesn’t guarantee you will not submit a duplicate. This check compares beneficiaries listed in the draft with any registrations you previously submitted during this registration period

This step gives attorneys a false sense of security. The USCIS duplicate check only looks at registrations submitted in the current registration period. It has no memory of previous seasons, no awareness of other batches submitted earlier in the same window. An attorney who uploaded two separate CSVs with overlapping beneficiaries will pass this check and still submit a duplicate. The tool tells you you're safe. It doesn't actually know.

11. H1B registration summary review
12. If satisfied, advance to the signatory page.
13. Pay and submit.
14. If the person is a legal representative - fill the form G28 and notify the company administrator, it needs to be done by a human.
15. Post admin step, user needs to click ‘Pay and Submit’
    This is the point of no return. USCIS does not have a cancellation flow. Once this is clicked, the registration is live, the fee is charged, and there is no undo.

Proposed Solution

No	Process Step	Action Taker	Rationale
1	Login to company account	**Human**	Security & liability - attorney credentials required
2	Navigate to 'File a Form' and select H-1B	**Human**	Workflow kickoff trigger - attorney initiates process
3a	Download latest USCIS CSV template	**Workflow**	Deterministic - scrape USCIS page for current template
3b	Validate data format (country codes, wage levels, field completeness)	**Workflow (LLM for edge cases)**	**Workflow Pydantic schema validation** - deterministic rules check all fields against USCIS template requirements. Catches 95%+ of errors instantly with zero cost. (LLM for Normalizing messy country names (like "United States" to "US") and for detecting gibberish in free-text fields
3c	Check for duplicate beneficiaries (across all firm data)	**Workflow**	**Fuzzy string matching (fuzzywuzzy/dedupe library)** using Levenshtein distance on

			name + exact match on passport/DOB.
3d	Verify entry count < 2500 per upload	**Workflow**	Simple row count validation
3e	**Flag validation errors for human review**	**Human Review**	Attorney corrects data issues before upload
4	Click through security reminder page	**Workflow + Human**	Workflow displays reminder, human acknowledges
5a	Upload CSV file to USCIS portal	**Human**	File uploading to be done by human
5b	OR individual form entry (if not using bulk upload)	**AI Agent (chained)**	Agent 1: Form filling. Agent 2: Field validation
6	Click 'Import' and parse page for errors	**Workflow + Agent + Human Review**	Workflow clicks, agent parses error messages, flags to human if errors detected
7	Verify green confirmation banner appears	**Workflow + Human Review**	**DOM element detection** - Playwright checks for success/error elements in page HTML using CSS selectors and text search. Fallback to screenshot for manual review if inconclusive
8	Download CSV copy of beneficiary table for records	**Workflow**	Automated file download and storage
9	Click 'Continue Registration' and enter registration info	**Workflow**	Navigation + form population from stored data
10	Fill authorized signatory information	**Workflow**	Pre-stored signatory details auto-populated
11	Run USCIS duplicate check (if individual entry method)	**Workflow + Human**	Workflow triggers check, attorney reviews results and confirms no duplicates
12	Review H-1B registration summary for data entry errors	**Agent + Human**	LLM performs semantic validation, flags inconsistencies, attorney performs final approval
13	Advance to signatory page (if satisfied)	**Workflow**	Navigation only after human approval

14	Pay registration fee and submit	**Human**	Financial info to be provided by attorney-
15	If legal representative: file Form G-28 and notify admin	**Human**	Professional responsibility - attorney signature required
16	Final 'Pay and Submit' action	**Human**	Final authorization - point of no return

What to fully automate

Step	What	Why Automate
3a	Download USCIS CSV template	Deterministic GET request to a known URL. Zero judgment required.
3b	Validate data format	Pydantic catches mismatched country codes, out-of-range wage levels, and missing required fields across hundreds of rows faster and more reliably than a human. Claude is only called edge cases like normalising "United States of America" to "US",etc.
3c	Duplicate detection	Fuzzy string matching on name plus exact match on passport and DOB runs across the firm's full beneficiary history in seconds. USCIS's own dedup tool only checks within a single draft — it won't catch "Jon Smith" filed last week against "John Smith" filed today.
3d	Entry count < 2500	One line of Python. Not worth a human's time to check manually.

8	Download confirmation CSV	Record-keeping that should never depend on a human remembering to do it. Playwright downloads it, S3 stores it.
9, 13	Navigation clicks	Just page advancement after the attorney has already approved the prior step. The decision was made in the gate before, this click is just executing it.
10	Fill signatory info	Removes friction that adds no value.

What to Partially Automate?

Step	What	Why hybrid
4	Security reminder page	The playwright navigates to it automatically.
6	Import and error parsing	Playwright clicks Import. Claude reads the error messages because USCIS error text is unstructured. Human fixes the data if errors are found.
7	Verify green banner	DOM check via CSS selector works 95% of the time. The 5% failure case is USCIS updating their UI and when the selector is inconclusive. Playwright takes a screenshot and blocks for human confirmation

11	USCIS duplicate check	The click is deterministic and Playwright handles it. Whether a name match is a genuine duplicate or a coincidence requires an attorney's judgment before proceeding.
12	Registration summary review	Claude catches the inconsistencies a schema check can't, like a wage level that doesn't match the job title. But the attorney makes the final call, because this is the last stop before an irreversible payment.

What’s manual? Why?

Step	What	Why Manual
1	Login	Automating login means storing attorney credentials server-side, a security risk Alma shouldn't own.
2	Kickoff navigation	The attorney consciously choosing to start a registration run is a feature.
3e	Error review and approval	The attorney reviews every flagged row and approves the corrected dataset. Fixing a bad registration after submission is far more expensive than fixing it here.
5a	CSV upload	The upload is the first action that touches USCIS, the attorney doing it manually is a deliberate second confirmation that the right file is going to the right place.

14	Payment	$215 per beneficiary, potentially hundreds of beneficiaries per run. The attorney must consciously authorise this expenditure.
15	Form G-28	If the attorney is filing on behalf of a client, they must sign and submit Form G-28 themselves and notify the company administrator, this cannot be delegated to the system.
16	Final Pay and Submit	Once clicked, this cannot be undone. USCIS has no cancellation flow.

Proposed Prototype : Attorney facing Dashboard

https://lottery-automation.lovable.app/

Key aspects taken into consideration while designing the prototype

1. **Primary user and their feelings** : The attorney is the primary user. They are looking for confidence and trust. They should feel certain that the right step/process is being executed.

2. **Important information that the primary user needs to always know** : they need to know the status of a batch in progress. For every batch, they need to know the stage of the completion process, they need to know if there is an additional review/input it needs from them and if anything has gone wrong.

3. **Information that needs to be displayed by human gate steps** : The respective step should show a screen that shows the specific data the attorney needs to review, a short summary of what the system checked and found and next steps to be executed by the attorney.

4. **Error state :** Error state needs to tell them three things - what happened, what they need to do, and whether any data was affected. Vague error messages in a legal context are not just bad UX, they are a liability**.**

5. **Success state :** After a successful registration, the portal should show a summary to the attorney with info such as how many beneficiaries were registered, confirmation numbers, etc. This screen should also have links to download the beneficiary info/status so they can share it with their client.

Prototype description

1. Dashboard shows three registration states -

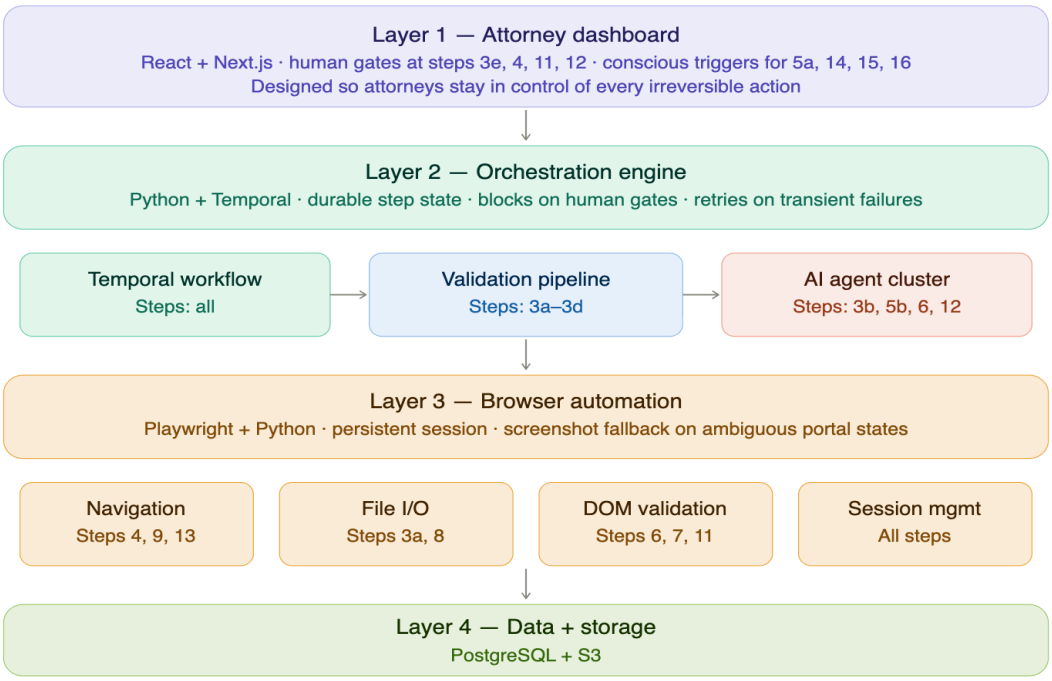
a. Action Required section - includes steps/ human gates that need attorney input to proceed further.

b. In Progress - steps that have the automation running and do not need attorney inputs.

c. Completed - confirmation screen with confirmation #, beneficiary count and the total amount paid and downloadable documents of successful registration.

2. Checkpoint pages guide attorneys through a human-in-the-loop workflow: Data Validation - Duplicate Check - Security Reminder - Payment - Final Review.

Technical Implementation



Layer	Component	Tools	What it does	Steps
1. Attorney dashboard	Human gates	React + Next.js	Blocks workflow at key decision points until attorney approves	3e, 4, 11, 12
1. Attorney dashboard	Conscious triggers	React + Next.js	Attorney manually initiates irreversible actions	5a, 14, 15, 16
2.Orchestration engine	Temporal workflow	Python + Temporal	Preserves full run state, parks on human gates, retries portal timeouts	All steps
2.Orchestration engine	Validation pipeline	Pydantic + Claude	Pydantic runs first for speed and cost. Claude only called for edge cases	3a–3d
2.Orchestration engine	AI agent cluster	Claude API	Fills forms, validates output, flags inconsistencies. Never auto-approves	3b, 5b, 6, 12
3. Browser automation	Navigation	Playwright + Python	Clicks through portal pages after human gates. Never clicks on behalf of human	4, 9, 13
3. Browser automation	File I/O	Playwright + Python	Fetches USCIS template, stores confirmation CSV. Upload stays human	3a, 8
3. Browser automation	DOM validation	Playwright + Python	Checks for green banner via CSS selector. Takes	6, 7, 11

			screenshot on ambiguous result	
3. Browser automation	Session mgmt	Playwright + Python	Persists cookies across portal session, pauses and alerts on CAPTCHA	All steps
4. Data + storage	PostgreSQL	PostgreSQL	Stores beneficiary history, signatory details, audit trail	All steps
4. Data + storage	S3	S3 compatible	Stores confirmation CSVs, screenshots, run artifacts timestamped per run	All steps

Known risks and failure handling

1. USCIS portal brittleness

The automation layer depends on reading the USCIS portal DOM via Playwright. USCIS may update their UI without notice, a selector that correctly identifies the confirmation banner today may break silently after a portal change, causing the workflow to stall or incorrectly report a successful upload.

Mitigation: DOM check runs first. If inconclusive, Playwright takes a screenshot and blocks for attorney confirmation before advancing. This prevents a false positive from propagating forward.

Limitation: This is a fallback, not a fix. The system will require active maintenance whenever USCIS updates their portal

2. Session timeouts

USCIS portal sessions expire after periods of inactivity. H-1B season is not a controlled environment - attorneys context-switch between clients, step away mid-run, and get pulled into calls. A session timeout mid-workflow is a likely occurrence, not an edge case.

Mitigation: Temporal preserves full workflow state at every step. On timeout detection, the workflow parks at the current step and surfaces a named alert on the attorney dashboard not a generic error, specifying where the run paused and confirming no data

was lost. The attorney re-authenticates and resumes from the exact step the workflow stopped.

Exception: A timeout during step 5a (CSV upload) cannot be automatically resolved. The system cannot confirm whether USCIS received the file before the session died. In this case the attorney is asked to check the USCIS portal directly and confirm one of two paths re-upload or continue, before the workflow resumes.

Success Metrics

	Metric	**Target Value**
1.	Time per Beneficiary Registration	<5 minutes (estimated 20 min baseline)
2.	Data Entry Error Rate	<0.5% (estimated 3% baseline)
3.	Duplicate Registration Incidents	0 per season
4.	Attorney Capacity	Beneficiaries per Attorney. (target 80-120 beneficiaries)
5.	Attorney Satisfaction Score	Need more data to determine a target value, but ideally >8.5/10
6	USCIS Registration Rejection Rate	Needs to be determined based on the baseline - current rejection rate (50% reduction in data-error rejections compared to the manual baseline)